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EXPERIMENT-12

Question

Write a **menu-driven C program using switch-case** to input a positive integer and perform one of the following operations:

1. Count the number of even digits and odd digits.
2. Find the sum of even digits and the sum of odd digits separately.

1. Algorithm

Step 1:Start.

Step 2:Input a positive integer n .

Step 3:Display the menu:

1. Count even and odd digits
2. Find sum of even and odd digits

Step 4:Input the user's choice.

Step 5:Use `switch` according to the choice.

Step 6: If choice is 1

- Set `even = 0` and `odd = 0`.
- Repeat while the number is greater than 0:
 - Find the last digit using `digit = n % 10`.
 - If `digit % 2 == 0`, increase `even` by 1.
 - Otherwise, increase `odd` by 1.
 - Remove the last digit using `n = n / 10`.
- Display the number of even and odd digits.

Step 7: If choice is 2

- Set `sum_even = 0` and `sum_odd = 0`.

- Repeat while the number is greater than 0:
 - Find the last digit using `digit = n % 10`.
 - If `digit % 2 == 0`, add the digit to `sum_even`.
 - Otherwise, add the digit to `sum_odd`.
 - Remove the last digit using `n = n / 10`.
- Display `sum_even` and `sum_odd`.

Step 8: If the choice is neither 1 nor 2, display **Invalid choice**.

Step 9: Stop.

2. Test Case Table

Test Case	Inputs	Expected Output	Actual Output	Results
1	Number = 12045, Choice = 1	Even digits = 3, Odd digits = 2	Even digits = 3, Odd digits = 2	Pass
2	Number = 12345, Choice = 2	Sum of even digits = 6, Sum of odd digits = 9	Sum of even digits = 6, Sum of odd digits = 9	Pass
3	Number = 2468, Choice = 1	Even digits = 4, Odd digits = 0	Even digits = 4, Odd digits = 0	Pass
4	Number = 1357, Choice = 2	Sum of even digits = 0, Sum of odd digits = 16	Sum of even digits = 0, Sum of odd digits = 16	Pass

3. C Program

```
#include <stdio.h>
int main()
{
    int n, choice, digit;
    int even = 0, odd = 0;
    int sum_even = 0, sum_odd = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\n1. Count even and odd digits");
    printf("\n2. Find sum of even and odd digits");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    switch(choice)
    {
        case 1:
            while(n > 0)
            {
                digit = n % 10;
```

```

        if(digit % 2 == 0)
            even++;
        else
            odd++;
        n = n / 10;
    }
    printf("Even digits = %d\n", even);
    printf("Odd digits = %d\n", odd);
    break;
case 2:
    while(n > 0)
    {
        digit = n % 10;
        if(digit % 2 == 0)
            sum_even = sum_even + digit;
        else
            sum_odd = sum_odd + digit;
        n = n / 10;
    }
    printf("Sum of even digits = %d\n", sum_even);
    printf("Sum of odd digits = %d\n", sum_odd);
    break;
default:
    printf("Invalid choice\n");
}
return 0;
}

```

4.Compilation and Execution

```
(kali㉿kali)-[~]  
$ gcc hdoc13.c -o hdoc13  
  
(kali㉿kali)-[~]  
$ ./hdoc13  
Enter a positive integer: 12045  
  
1. Count even and odd digits  
2. Find sum of even and odd digits  
Enter your choice: 1  
Even digits = 3  
Odd digits = 2
```

```
(kali㉿kali)-[~]  
$ gcc hdoc13.c -o hdoc13  
  
(kali㉿kali)-[~]  
$ ./hdoc13  
Enter a positive integer: 12345  
  
1. Count even and odd digits  
2. Find sum of even and odd digits  
Enter your choice: 2  
Sum of even digits = 6  
Sum of odd digits = 9
```

```
(kali㉿kali)-[~]  
$ gcc hdoc13.c -o hdoc13  
  
(kali㉿kali)-[~]  
$ ./hdoc13  
Enter a positive integer: 2468  
  
1. Count even and odd digits  
2. Find sum of even and odd digits  
Enter your choice: 1  
Even digits = 4  
Odd digits = 0
```

```
(kali㉿kali)-[~]  
$ gcc hdoc13.c -o hdoc13  
  
(kali㉿kali)-[~]  
$ ./hdoc13  
Enter a positive integer: 1357  
  
1. Count even and odd digits  
2. Find sum of even and odd digits  
Enter your choice: 2  
Sum of even digits = 0  
Sum of odd digits = 16
```

5.Testing Against Test Cases

Test Case 1

Input: Number = 12045, Choice = 1

Expected Output: Even digits = 3, Odd digits = 2

Actual Output: Even digits = 3, Odd digits = 2

Result: **Pass**

Test Case 2

Input: Number = 12345, Choice = 2

Expected Output: Sum of even digits = 6, Sum of odd digits = 9

Actual Output: Sum of even digits = 6, Sum of odd digits = 9
Result: **Pass**